Parity under standard-scaling neural-network Gaussian processes: superpolynomial sample lower bounds at fixed depth, and an obstruction

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Abstract

We study Bayesian learning of parity by fully connected networks with the standard neural-network Gaussian-process (NNGP) prior scaling: weight variances $\sigma_1^2/(d+1)$ in the first layer and σ_ℓ^2/N in later layers, at fixed depth, with width $N \to \infty$ first. The prior is conditioned on an output margin $|z(x)| > a$ at every Boolean input and then on the training signs; prediction counts as correct only when the correct label has posterior probability greater than $2/3$. For tanh networks of every fixed depth and every choice of layer gains, we prove that, for all large d, $n \le \exp(\frac{1}{20}\sqrt{d}\log d)$ uniform independent examples of full parity on d bits leave, with probability $1 - e^{-\Omega(d)}$ over the sample, at least a $1/16$ fraction of inputs without confident correct prediction. For the learner without the global margin, conditioned only on the training signs, a threshold of the same form $d^{\Theta(\sqrt{d})}$ (explicitly $d^{\sqrt{d}/10-1/2}/6000$ for $d \ge 25$) holds on every training set, for every bounded continuous nonconstant activation, every depth and every gain; the argument is kernel-agnostic and also gives the scale d^k for fixed k-sparse parity. An $\exp(\Omega(d))$ lower bound, the analogue of the subcritical theorem in this collection, remains open. We prove that the step of that theorem which bounds the posterior's conditioning cost by a polynomial per sample is false at the standard scaling: for tanh networks and $\exp(d^\epsilon) \le n \le e^{cd}$, the cost per sample is superpolynomial in d with high probability. A matching upper bound on that cost would complete an exponential proof. Numerical spectra and small-dimensional posterior computations illustrate the results.

1 Introduction and results

The collection's subcritical theorem [1] proves an $\exp(\Omega(d))$ sample lower bound for full parity under a family of Gaussian network priors whose width scaling is indexed by an exponent $\gamma < 1$. In that family the hidden preactivations diverge with width, tanh acts as a sign function, and the limiting covariance is the iterated arcsine kernel. The usual NNGP scaling is different: it keeps the preactivations of order one, so the limit is the dual-activation kernel of tanh itself. In the exponent notation of the collection's FAQ (Question 9) this scaling is the tuple $(0, 1, \dots, 1, 1)$, which is not on the one-parameter line $(\gamma - 1, \gamma, \dots, \gamma, \gamma + 1)$. The collection's NNGP note [2] proves a $\Theta(2^d)$ bound for a tanh NNGP whose depth grows linearly with d, and states that the fixed-depth problem is open.

We ask whether parity needs exponentially many samples for the fixed-depth standard-scaling learner. The answer we can prove is partial.

Results. All statements take width to infinity first, at fixed dimension, depth, gains, margin and training sample.

1. **Width limits** (Proposition 2.4, proved). For every bounded continuous nonconstant activation the network output converges to a nondegenerate Gaussian process with a dot-product kernel $K = A\,F((1 + x \cdot y)/(d + 1))$, where F has nonnegative Taylor coefficients that do not depend on d. Both learners' posterior label probabilities converge to those of the conditioned Gaussian process.

2. **Sign-only learner, every training set** (Theorem 3.2 and Corollary 3.4, proved). Without the global margin, for every such activation, every depth and every gain, every training set of size $n \le d^{\sqrt{d}/10-1/2}/6000$ (with $d \ge 25$; the explicit bound exceeds one only for $d \ge 385$) leaves, in the width limit, at least a quarter of the inputs of full parity without confidence above $2/3$. For k-sparse parity the threshold is $\binom{d}{\lfloor\sqrt{k}/5\rfloor}/6000$, and for fixed k the failure fraction is at least $2^{-k} - O(n/\binom{d}{k})$. The same bounds hold for the margin learner when the readout variance is $\sigma_L^2 N^{-b}$ with $b < 1$ (Remark 3.5).

3. **Margin learner, typical samples** (Theorem 5.5, proved). For tanh at every fixed depth and every choice of gains and margin, $n \le \exp(\frac{1}{20}\sqrt{d}\log d)$ iid uniform examples of full parity leave, for $d \ge d_0$ (not explicit) and with probability at least $1 - e^{-d/10}$, Gibbs error at least $3/8$ and confidence failure at least $1/16$ in the width limit.

4. **The exponential question** (open). We do not know whether $\exp(\Omega(d))$ samples are needed. Proposition 6.2 (proved) shows that one specific step of the subcritical proof fails. That step bounds the conditioning cost per sample by a polynomial in d (at most $Cd^{1+\beta}$). Here, for $\exp(d^\epsilon) \le n \le e^{c_1 d}$ and with probability at least $1 - 2\delta$, that cost is superpolynomial in d. We also identify a matching upper bound on that cost which would complete an $\exp(\Omega(d))$ proof (Section 6).

The sign-only result uses only a universal inequality between consecutive level masses of an isotropic kernel (Lemma 2.6). The margin result additionally needs a lower bound on every eigenvalue of the kernel. We derive it from an explicit Fourier-integral formula for the Hermite coefficients of tanh (Lemma 4.1).

What this paper does not show. We prove no $\exp(\Omega(d))$ lower bound and no upper bound for either learner. We do not decide whether the exponent $\sqrt{d}\log d$ is the truth. The margin theorem is proved for tanh, and for odd bounded continuous activations whose first-layer dual coefficients are all positive and decay at most like $e^{-o(\sqrt{m}\log m)}$ (Remark 5.6). It holds for $d \ge d_0$, where d_0 depends on the depth, gains and margin and is not explicit, because we prove no rate for the coefficient tail $T(s)$. It does not cover erf, whose coefficients decay geometrically. As in [2, 1], the results concern the exact posterior; nothing is claimed about computing it.

Relation to kernel regression. The NNGP posterior mean given real-valued labels is kernel interpolation. For it, the span argument of [2, Proposition 4.1] and the lower bounds of Hsu and of Simon et al. for kernel regression concern squared error or correlation of the mean. Our learners condition on signs, or on signs and a global margin, and are judged by posterior confidence. That is a different criterion ([2, Section 5]; FAQ Question 7).

2 Model, width limits, and the cube spectrum

2.1 Network and learners

Let $\mathcal{X} = \{-1,1\}^d$ with the uniform probability measure, $M = 2^d$ and $D = d+1$. Fix a depth $L \geq 2$, an activation $\phi : \mathbb{R} \to \mathbb{R}$, gains $\sigma_1, \ldots, \sigma_L > 0$, and a margin $a > 0$. For width N put

$$h^0(x) = (1, x) \in \mathbb{R}^D, \qquad h^\ell(x) = \phi(W^\ell h^{\ell-1}(x)) \ (1 \leq \ell \leq L-1), \qquad z_N(x) = W^L h^{L-1}(x),$$

with independent centered Gaussian weights of variances

$$\operatorname{Var} W^1_{ij} = \frac{\sigma_1^2}{D}, \qquad \operatorname{Var} W^\ell_{ij} = \frac{\sigma_\ell^2}{N} \ (2 \leq \ell \leq L-1), \qquad \operatorname{Var} W^L_i = \frac{\sigma_L^2}{N}. \tag{1}$$

All hidden layers have width N, and there are no biases besides the constant input coordinate. The *unit-gain* network has every $\sigma_\ell = 1$. The specified-gain family of [2, Section 7] has $\sigma_1 = 1$ and $\sigma_\ell^2 = 1/\mathbb{E}\tanh^2(G)$ for $\ell \geq 2$, with $G \sim N(0,1)$; at fixed depth it is another instance of (1).

Assumption 2.1. ϕ is bounded, continuous and not constant.

Let $f : \mathcal{X} \to \{-1,1\}$ be a target and $S = (X_1, \ldots, X_n)$ a sequence of training inputs. Define the events

$$B_S^{\mathrm{mar}} = \{z : \ |z(x)| > a \ \forall x \in \mathcal{X}, \ f(X_i)z(X_i) > 0 \ \forall i\},$$
$$B_S^{\mathrm{sgn}} = \{z : \ f(X_i)z(X_i) > 0 \ \forall i\}.$$

The *margin learner* is the prior conditioned on B_S^{mar}; this is the collection's convention (FAQ Question 1). The *sign-only learner* is the prior conditioned on B_S^{sgn}. For either learner put

$$q_{N,S}(x) = \mathbb{P}\{f(x)z_N(x) > 0 \mid z_N \in B_S\},$$
$$F_N(S) = \mathbb{P}_x\{q_{N,S}(x) \leq 2/3\}, \qquad R_N(S) = \mathbb{E}_x[1 - q_{N,S}(x)].$$

F_N is the confidence-failure fraction and R_N the Gibbs error. For every posterior and every N,

$$R_N \leq \frac{1 - F_N}{3} + F_N, \qquad \text{so} \qquad F_N \geq \frac{3R_N - 1}{2}, \tag{2}$$

since $1 - q \leq 1$ on the failure set and $1 - q < 1/3$ off it.

2.2 Dual activations

For $q > 0$ write $v_\phi(q) = \mathbb{E}\phi(\sqrt{q}G)^2$ and, for $\rho \in [-1,1]$,

$$\widehat{\phi}_q(\rho) = \frac{\mathbb{E}\phi(\sqrt{q}U)\phi(\sqrt{q}V)}{v_\phi(q)},$$

where (U,V) are standard Gaussians with correlation ρ. Under Assumption 2.1, $v_\phi(q) > 0$.

Lemma 2.2 (Dual activations)**.** *Expand $\phi(\sqrt{q}g) = \sum_{m\geq 0} \beta_m He_m(g)/\sqrt{m!}$ in $L^2(N(0,1))$, with probabilists' Hermite polynomials He_m. Then*

$$\widehat{\phi}_q(\rho) = \sum_{m\geq 0} c_m(q)\rho^m, \qquad c_m(q) = \frac{\beta_m^2}{\sum_k \beta_k^2} \geq 0, \qquad \sum_m c_m(q) = 1,$$

with absolute convergence on $[-1,1]$. If ϕ is odd, $c_m(q) = 0$ for even m. Under Assumption 2.1, infinitely many $c_m(q)$ are positive.

Proof. For correlated standard Gaussians, $\mathbb{E}He_m(U)He_k(V) = \delta_{mk}\, m!\, \rho^m$; compare coefficients of $s^m t^k$ in $\mathbb{E}e^{sU-s^2/2}e^{tV-t^2/2} = e^{\rho st}$. The bilinear expansion of $\mathbb{E}\phi(\sqrt{q}U)\phi(\sqrt{q}V)$ then converges absolutely, because $|\rho| \le 1$ and $\sum \beta_m^2 = v_\phi(q) < \infty$. Odd functions have no even Hermite components. A bounded nonconstant function is not a polynomial, so infinitely many β_m are nonzero. □

Define $q_1 = \sigma_1^2$, $v_\ell = v_\phi(q_\ell)$, $q_{\ell+1} = \sigma_{\ell+1}^2 v_\ell$ for $1 \le \ell \le L-2$, and

$$A = \sigma_L^2 v_{L-1}, \qquad F = \widehat{\phi}_{q_{L-1}} \circ \cdots \circ \widehat{\phi}_{q_1}, \qquad K(x,y) = A\, F\!\left(\frac{1 + x \cdot y}{D}\right). \tag{3}$$

Lemma 2.3. *$F(t) = \sum_{m \ge 0} c_m t^m$ with $c_m \ge 0$ and $\sum_m c_m = 1$. The coefficients c_m and the scale A depend on ϕ, L and the gains, but not on d. If ϕ is odd, F is odd. Under Assumption 2.1, infinitely many c_m are positive.*

Proof. Composition of power series with nonnegative coefficients summing to one preserves both properties, and composition of odd series is odd. The variances q_ℓ do not involve d, because $\|h^0(x)\|^2 = D$ for every input. For the last claim, induct on the number of layers. Suppose H has infinitely many positive coefficients and $H(1) = 1$. Some $c_k(q)$ with $k \ge 1$ is positive; otherwise $\widehat{\phi}_q$ would be the constant $c_0(q) = 1$, forcing $\beta_m = 0$ for $m \ge 1$ and ϕ constant on the support of a Gaussian. The series H^{k-1} has some positive coefficient, at degree e say, so $H^k = H \cdot H^{k-1}$ has a positive coefficient at degree $e + m$ for each of the infinitely many m with $[t^m]H > 0$. □

2.3 Width limits

Proposition 2.4 (Width limits)**.** *Under Assumption 2.1, fix d, L and the gains.*

(i) As $N \to \infty$, z_N converges in law on $\mathbb{R}^{\mathcal{X}}$ to $G \sim N(0, K)$, with K as in (3).

(ii) K is positive definite (Lemma 2.5).

(iii) For every target f, training sequence S and margin $a > 0$, and for either learner, $\mathbb{P}\{z_N \in B_S\} \to \mathbb{P}\{G \in B_S\} > 0$, and $q_{N,S}(x) \to q_S(x) := \mathbb{P}\{f(x)G(x) > 0 \mid G \in B_S\}$ for every x. Consequently $R_N(S) \to R_S := \mathbb{E}_x[1 - q_S(x)]$, $\liminf_N F_N(S) \ge (3R_S - 1)/2$, and $\liminf_N F_N(S) \ge \mathbb{P}_x\{q_S(x) < 2/3\}$.

Proof. (i) For a positive semidefinite matrix Σ on $\mathcal{X}$ let $\Gamma(\Sigma) = \mathbb{E}[\phi(u)\phi(u)^\mathsf{T}]$ with $u \sim N(0, \Sigma)$. The map Γ is continuous on the positive semidefinite cone: if $\Sigma_k \to \Sigma$, then $\Sigma_k^{1/2}\xi \to \Sigma^{1/2}\xi$ for every ξ, and dominated convergence applies because ϕ is bounded and continuous. Write $Q_N^\ell(x,y) = N^{-1}\sum_i h_i^\ell(x) h_i^\ell(y)$ and let $\mathcal{F}_\ell$ be generated by the first ℓ layers of weights. The first-layer rows are independent, and each row's preactivation vector is $N(0, \Sigma_1)$ with $\Sigma_1(x,y) = \sigma_1^2(1 + x\cdot y)/D$. Conditionally on $\mathcal{F}_{\ell-1}$ with $\ell \ge 2$, the rows of the ℓth layer are independent with preactivation law $N(0, \sigma_\ell^2 Q_N^{\ell-1})$. Hence $\mathbb{E}[Q_N^\ell \mid \mathcal{F}_{\ell-1}] = \Gamma(\sigma_\ell^2 Q_N^{\ell-1})$ (with $\Gamma(\Sigma_1)$ at $\ell = 1$), and every entry has conditional variance at most $\|\phi\|_\infty^4/N$. By Chebyshev's inequality and the continuous mapping theorem, induction on ℓ gives $Q_N^\ell \to K^\ell$ in probability, where $K^1 = \Gamma(\Sigma_1)$ and $K^\ell = \Gamma(\sigma_\ell^2 K^{\ell-1})$. Every diagonal entry of $\sigma_\ell^2 K^{\ell-1}$ equals q_ℓ, and the correlations follow the recursion $\rho_1 = (1 + x\cdot y)/D$, $\rho_{\ell+1} = \widehat{\phi}_{q_\ell}(\rho_\ell)$. Therefore $\sigma_L^2 K^{L-1} = K$. Finally $z_N \mid \mathcal{F}_{L-1} \sim N(0, \sigma_L^2 Q_N^{L-1})$, so for each $t \in \mathbb{R}^{\mathcal{X}}$ the characteristic function is $\mathbb{E}\exp(-\frac{1}{2}\sigma_L^2 t^\mathsf{T} Q_N^{L-1} t)$, which converges to $\exp(-\frac{1}{2}t^\mathsf{T} K t)$ by bounded convergence.

(iii) Every coordinate $G(x)$ has variance $A > 0$. The boundary of B_S, and of $B_S \cap \{f(x)z(x) > 0\}$, lies in the finite union of hyperplanes $\{z(y) = \pm a\}$ and $\{z(y) = 0\}$, which is G-null. By the portmanteau theorem these probabilities converge. The limit of the denominator is positive because

K is nonsingular and B_S is a nonempty open set. The last assertions follow from (2), from the finiteness of $\mathcal{X}$, and from the fact that $q_S(x) < 2/3$ forces $q_{N,S}(x) < 2/3$ for all large N. □

The quantifier order is the one in the main results: $N \to \infty$ at fixed d, L, gains, a and S. No quantitative width is claimed.

2.4 The kernel on the cube

For $V \subseteq \{0, 1, \dots, d\}$ write $\chi_V(u) = \prod_{i \in V} u_i$ on $\{-1, 1\}^D$, and χ_U for $U \subseteq [d]$ on $\mathcal{X}$. Let $p_m(j)$ be the probability that the walk on $\{-1, 1\}^D$ which flips one uniformly chosen coordinate per step is, after m steps, at a specified vertex at distance j from its start.

Lemma 2.5 (Cube spectrum; the kernel-free part of [1, Lemma 3.1]). *Let $F = \sum_m c_m t^m$ with $c_m \geq 0$ and $\sum c_m = 1$. Then $F(u \cdot u'/D) = \sum_V a_{|V|} \chi_V(u) \chi_V(u')$ on $\{-1, 1\}^D$, with $a_j = \sum_m c_m p_m(j)$, and:*

(a) $a_j \geq a_{j+2}$ for $0 \leq j \leq D - 2$;

(b) $a_j > 0$ if and only if $c_m > 0$ for some $m \geq j$ with $m \equiv j \pmod 2$;

(c) on $\mathcal{X}$, $K(x, y) = \sum_{U \subseteq [d]} \lambda_{|U|} \chi_U(x) \chi_U(y)$ with $\lambda_q = A(a_q + a_{q+1})$, and λ_q is nonincreasing in q;

(d) if infinitely many c_m are positive, every λ_q is positive and K is positive definite;

(e) if F is odd, $\lambda_{2k} = \lambda_{2k+1} = A a_{2k+1}$.

Proof. Expanding $(u \cdot u'/D)^m$ as an average over m independent uniform coordinates shows that its coefficient of $\chi_V(u)\chi_V(u')$ is $p_m(|V|)$. For (a), the one-step recursion $p_{m+1}(j) = \frac{j}{D} p_m(j-1) + \frac{D-j}{D} p_m(j+1)$ gives

$$D[p_{m+1}(j) - p_{m+1}(j+2)] = j[p_m(j-1) - p_m(j+1)] + (D-j-2)[p_m(j+1) - p_m(j+3)],$$

with $p_m(i) := 0$ for $i \notin [0, D]$, and induction from $m = 0$ gives $p_m(j) \geq p_m(j+2)$ (the case $j + 1 = D - 1$ of the hypothesis is $p_m(D-1) \geq p_m(D+1) = 0$, which is trivial); average against $c_m \geq 0$. For (b), $p_m(j) > 0$ exactly when $m \geq j$ and $m \equiv j$. For (c), restrict to $u = (1, x)$: $\chi_V(1, x) = \chi_{V \setminus \{0\}}(x)$, and each $U \subseteq [d]$ arises from $V = U$ and $V = U \cup \{0\}$. Monotonicity follows from (a): $\lambda_{q+1}/A = a_{q+1} + a_{q+2} \leq a_{q+1} + a_q$. For (d), given q there is $m \geq q + 1$ with $c_m > 0$, and by (b) one of a_q, a_{q+1} (the one whose index has the parity of m) is positive. The matrix K has eigenvalue $M\lambda_{|U|}$ on χ_U. Part (e) is (c) with $a_{\text{even}} = 0$. □

Throughout, λ_q denotes these *normalized* eigenvalues of K itself, so they include the factor A. For $g : \mathcal{X} \to \mathbb{R}$ write $\widehat{g}(U) = \mathbb{E}_x g(x)\chi_U(x)$, $\|g\|_2^2 = \mathbb{E}_x g(x)^2$, $g_{\leq r} = \sum_{|U| \leq r} \widehat{g}(U)\chi_U$ and $g_{>r} = g - g_{\leq r}$. The Gaussian energy is $\|g\|_K^2 = g^{\mathsf{T}} K^{-1} g = \sum_U \widehat{g}(U)^2/\lambda_{|U|}$, the convention of [1]. Let

$$\tau_r = \sum_{q > r} \binom{d}{q} \lambda_q = \operatorname{Var} G_{>r}(x), \qquad T(r) = \sum_{m > r} c_m.$$

Lemma 2.6 (Level-mass inequalities). *Let F be as in Lemma 2.5 and $0 \leq r < d$.*

(a) $\binom{d}{r+1} \lambda_{r+1} \leq \tau_r$.

(b) If F is odd and r is odd, $\binom{D}{r+2} \lambda_{r+1} \leq \tau_r$.

(c) $\tau_r \le A\,T(r)$.

(d) $s_{2,r} := \sum_{q>r} \binom{d}{q}\lambda_q^2 \le \lambda_{r+1}\tau_r$.

Proof. (a) τ_r contains the term $q = r+1$. (b) Here $\lambda_{r+1} = \lambda_{r+2}$ by Lemma 2.5(e), and $\binom{d}{r+1} + \binom{d}{r+2} = \binom{D}{r+2}$. (c) By the grouping in Lemma 2.5(c), $\tau_r/A = \sum_{V:|V\setminus\{0\}|>r} a_{|V|} \le \sum_{j>r}\binom{D}{j}a_j$, and $\binom{D}{j}a_j = \sum_m c_m \mathbb{P}\{\text{walk at distance } j \text{ at time } m\}$. Distance j at time m requires $m \ge j$, so $\sum_{j>r}\binom{D}{j}a_j \le \sum_{m>r} c_m$. (d) Bound one factor λ_q by λ_{r+1}, using monotonicity. □

Inequality (a) is what makes the sign-only argument kernel-agnostic. The subcritical paper uses the sharper arcsine ratio $\lambda_{r+1}/\tau_r \asymp 1/(r\binom{D}{r+2})$ [1, Lemma 3.2]; we use only (a) and (b), which hold for every such kernel.

3 The sign-only learner on every training set

In this section K is any positive-definite covariance on $\mathcal{X}$ that depends only on Hamming distance and whose degree eigenvalues λ_q are nonincreasing. By Lemma 2.5 this includes every kernel (3) under Assumption 2.1, for every depth and gain. Let $G \sim N(0, K)$ and

$$q_S(x) = \mathbb{P}\{f(x)G(x) > 0 \mid f(X_i)G(X_i) > 0\ \forall i\}.$$

For $0 \le r < d$ let

$$\mathrm{err}_r(f) = \min_{\deg p \le r} \mathbb{P}_x\{f(x)p(x) \le 0\},$$

the least error of a degree-r polynomial threshold function with zeros counted as errors. A small constant shift of p removes its zeros without creating new errors, so this agrees with the definition used in [3].

Lemma 3.1 (Gaussian mean and confidence [1, Lemma 10.1])**.** *Let $\theta = 13/20$ and $c_\theta = \sqrt{2\pi}(e^{-1} - 7/20)$. Let $m = \mathbb{E}[G \mid f(X_i)G(X_i) > 0\ \forall i]$, let $s \le n$ be the number of distinct training inputs, and let $\sigma_x^2 = \mathrm{Var}(G(x) \mid G_S)$. Then for every training sequence and every $0 \le v < d$:*

(i) $\|m\|_K^2 \le s$;

(ii) if $\sigma_x > 0$ *and* $q_S(x) > \theta$, *then* $f(x)m(x) > c_\theta\sigma_x$;

(iii) $\mathbb{P}_x\{\sigma_x^2 < \tau_v/2\} \le 2s\lambda_{v+1}/\tau_v \le 2n\lambda_{v+1}/\tau_v$;

(iv) $\|m - m_{\le v}\|_2^2 \le n\lambda_{v+1}$.

Proof. This is [1, Lemma 10.1]. Its proof uses Gaussianity, positive definiteness, Prékopa's theorem and Fourier orthogonality. It uses the kernel only through two facts: the high-degree part $G_{>v}$ is independent of $G_{\le v}$ and has constant variance τ_v, and the largest normalized eigenvalue above degree v is λ_{v+1}. The second fact holds here by monotonicity. For (iii): conditioning also on $G_{\le v}$ can only lower the conditional variance, so $\sigma_x^2 \ge \tau_v - R(x,x)$, where $R = Q_{\mathcal{X}S}Q_{SS}^{\dagger}Q_{S\mathcal{X}}$ and Q is the covariance of $G_{>v}$. R is positive semidefinite of rank at most s, with $\mathrm{tr}\,R \le s\|Q\|_{\mathrm{op}} = sM\lambda_{v+1}$. Markov's inequality then gives (iii). Part (iv) follows from (i), since $\|m_{>v}\|_2^2 = \sum_{|U|>v}\widehat{m}(U)^2 \le \lambda_{v+1}\|m\|_K^2$. □

Theorem 3.2 (Sign-only learner)**.** *For every target f, every sequence S of n inputs (repeats and deliberately chosen inputs allowed), and every $0 \le r < d$,*

$$\mathbb{P}_x\{q_S(x) \le 13/20\} \;\ge\; \mathrm{err}_r(f) - C_0\frac{n\lambda_{r+1}}{\tau_r} \;\ge\; \mathrm{err}_r(f) - \frac{C_0\, n}{\binom{d}{r+1}}, \qquad C_0 = 2 + \frac{2}{c_\theta^2} < 1000. \quad (4)$$

If the eigenvalues are paired as in Lemma 2.5(e) and r is odd, $\binom{d}{r+1}$ may be replaced by $\binom{d+1}{r+2}$.

Proof. Let $p = m_{\le r}$. Let $E_1 = \{\sigma_x^2 < \tau_r/2\}$ and $E_2 = \{|m - p| \ge c_\theta\sqrt{\tau_r/2}\}$. By Lemma 3.1(iii), (iv) and Markov's inequality, $\mathbb{P}(E_1) + \mathbb{P}(E_2) \le (2 + 2/c_\theta^2)n\lambda_{r+1}/\tau_r$. Outside $E_1 \cup E_2$, the inequality $q_S(x) > 13/20$ implies $f(x)m(x) > c_\theta\sigma_x \ge c_\theta\sqrt{\tau_r/2} > |m(x) - p(x)|$, hence $f(x)p(x) > 0$. Therefore $\{fp \le 0\} \subseteq \{q_S \le 13/20\} \cup E_1 \cup E_2$. Since p has degree at most r, $\mathbb{P}_x\{fp \le 0\} \ge \mathrm{err}_r(f)$. The second inequality in (4) is Lemma 2.6(a), and the paired version is Lemma 2.6(b). Numerically $c_\theta = 0.04481\ldots$, so $C_0 = 997.7\ldots$. □

Lemma 3.3 (Parity is far from low-degree thresholds)**.** *Let $\chi_k(x) = x_1 \cdots x_k$ with $1 \le k \le d$, and let $0 \le s < k$. Then*

$$\mathrm{err}_s(\chi_k) \;\ge\; 2^{-k} \sum_{j \le (k-s-1)/2} \binom{k}{j} \;\ge\; \max\Big\{2^{-k},\; \frac{1}{2} - \frac{s+1}{\sqrt{2\pi k}}\Big\}.$$

Proof. On the k-cube the first inequality is the Aspnes–Beigel–Furst–Rudich bound [4], proved in [1, Lemma 6.2]. For $d > k$, fixing the coordinates outside $[k]$ turns a degree-s polynomial into a degree-s polynomial on the k-cube; average over the fixed coordinates. The term $j = 0$ gives 2^{-k}. Write $t = \lfloor (k - s - 1)/2 \rfloor$ and $B \sim \mathrm{Bin}(k, 1/2)$. By symmetry, $\mathbb{P}\{B \le t\} = \frac{1}{2}(1 - \mathbb{P}\{t < B < k - t\})$. The middle range contains $k - 2t - 1 \le s + 1$ integers, each of probability at most $\sqrt{2/(\pi k)}$. □

Corollary 3.4 (Networks without the global margin)**.** *Under Assumption 2.1, for every depth $L \ge 2$ and all gains, the sign-only network learner satisfies, for every target f, every training sequence S of length n and every $0 \le r < d$,*

$$\liminf_{N\to\infty} F_N(S) \;\ge\; \mathrm{err}_r(f) - \frac{C_0\, n}{\binom{d}{r+1}}.$$

In particular:

- *(a)* Full parity *($f = \chi_d$, $d \ge 25$): every training sequence with $n \le \binom{d}{\lfloor\sqrt{d}/5\rfloor}/(6C_0)$, and hence every one with $n \le d^{\sqrt{d}/10 - 1/2}/6000$, has $\liminf_N F_N(S) \ge 1/4$.*
- *(b)* Sparse parity *($f = \chi_k$, $25 \le k \le d$): every training sequence with $n \le \binom{d}{\lfloor\sqrt{k}/5\rfloor}/(6C_0)$ has $\liminf_N F_N(S) \ge 1/4$.*
- *(c)* Fixed k: $\liminf_N F_N(S) \ge 2^{-k} - C_0 n/\binom{d}{k}$ for every training sequence.*

These constants do not depend on L, the gains, or ϕ.

Proof. Combine Proposition 2.4(iii), the inclusion $\{q_S \le 13/20\} \subseteq \{q_S < 2/3\}$, and Theorem 3.2. For (a) and (b) take $r + 1 = \lfloor\sqrt{k}/5\rfloor$ (with $k = d$ in (a)). Lemma 3.3 gives $\mathrm{err}_r \ge \frac{1}{2} - \frac{1}{5\sqrt{2\pi}} > 0.42$, and the sample bound makes the subtracted term at most $1/6$. For the explicit form in (a), $\binom{d}{t} \ge (d/t)^t$ with $t = \lfloor\sqrt{d}/5\rfloor \ge \sqrt{d}/5 - 1$, so $\log\binom{d}{t} \ge (\sqrt{d}/5 - 1)\log(5\sqrt{d}) \ge \frac{\sqrt{d}}{10}\log d - \frac{1}{2}\log d$. For (c) take $r = k - 1$ and use $\mathrm{err}_{k-1}(\chi_k) \ge 2^{-k}$. □

Remark 3.5 (Larger readout variance). Corollary 3.4 also covers the *margin* learner when the readout variance is $\sigma_L^2 N^{-b}$ with $b < 1$, all other variances being as in (1). This is the tuple $(0, 1, \dots, 1, b)$ in the notation of FAQ Question 9. Indeed $z_N = N^{(1-b)/2}\tilde{z}_N$, where $\tilde{z}_N$ is the output at $b = 1$. So the margin event is $\{|\tilde{z}_N(x)| > \varepsilon_N \ \forall x\}$ with $\varepsilon_N = aN^{-(1-b)/2} \to 0$. For fixed $\varepsilon > 0$ and large N, $B_\varepsilon \subseteq B_{\varepsilon_N} \subseteq B_S^{\mathrm{sgn}}$, where B_ε adds the condition $|z| > \varepsilon$ everywhere to the training signs. Proposition 2.4(i) and the portmanteau theorem bound the liminf and limsup of $\mathbb{P}\{\tilde{z}_N \in B_{\varepsilon_N}\}$ by $\mathbb{P}\{G \in B_\varepsilon\}$ and $\mathbb{P}\{G \in B_S^{\mathrm{sgn}}\}$. Letting $\varepsilon \downarrow 0$ shows that these agree in the limit, since $\mathbb{P}\{G(x) = 0 \text{ for some } x\} = 0$. The same holds with the extra event $\{f(x)z(x) > 0\}$. So $q_{N,S}(x)$ converges to the sign-only $q_S(x)$, as in the $\gamma < 0$ case of [1, Proposition 2.2].

Remark 3.6. (1) The bound holds on *every* training set, including deliberately chosen ones, and uniformly in depth, gain and activation. Depth may even grow with d. For comparison, [2, Theorem 7.1] proves a $\Theta(2^d)$ bound for a different learner, the margin learner with specified gains, at depth at least $C_a(d+1)$. (2) Theorem 3.2 is [3, Proposition 3] with the arcsine eigenvalue ratio replaced by Lemma 2.6(a). (3) For the arcsine kernel the subcritical paper improves $\sqrt{d}\log d$ to d, for typical samples [1, Theorem 10.2]. That argument needs the sharp ratio $\lambda_{r+1}/\tau_r \asymp 1/(r\binom{D}{r+2})$ at r linear in d, and regular decay of the level masses $\binom{d}{q}\lambda_q$. For tanh the level masses decay much faster (Section 7), and we have not proved the matching estimates. The scale $\sqrt{d}\log d$ comes from requiring $\mathrm{err}_r(f)$ to stay bounded below (the Aspnes–Beigel–Furst–Rudich lemma); with Lemma 2.6 alone, larger r yields only a polynomially small failure fraction. For odd activations the paired form improves the binomial factor, for example to $\binom{d+1}{k+1}$ in Corollary 3.4(c) when k is even, without changing the exponent in (a) and (b).

4 Hermite coefficients of tanh and eigenvalue lower bounds

Lemma 4.1 (Hermite coefficients of tanh)**.** *Let $\sigma > 0$, $v = \mathbb{E}\tanh^2(\sigma G)$, and for odd m*

$$I_m(\sigma) = \int_0^\infty \frac{\omega^m e^{-\sigma^2\omega^2/2}}{\sinh(\pi\omega/2)}\, d\omega.$$

The dual coefficients of $\phi = \tanh$ at variance σ^2 are $c_m(\sigma^2) = \sigma^{2m} I_m(\sigma)^2/(m!\, v)$ for odd m and 0 for even m. In particular every odd coefficient is positive, and

$$c_m(\sigma^2) \ \geq \ \frac{2\sqrt{2\pi}}{\sigma^2 v\sqrt{m+1}}\, e^{-\pi\sqrt{m+1}/\sigma} \qquad (m \text{ odd}). \tag{5}$$

Proof. Gaussian integration by parts gives $\mathbb{E}[g(G)He_m(G)] = \mathbb{E}g^{(m)}(G)$ for smooth g with bounded derivatives. With $g(t) = \tanh(\sigma t)$ this gives $\beta_m = \sigma^m \mathbb{E}[\tanh^{(m)}(\sigma G)]/\sqrt{m!}$. The function $\mathrm{sech}^2 = \tanh'$ has Fourier transform $\int_{\mathbb{R}} \mathrm{sech}^2(y) e^{-i\omega y} dy = \pi\omega/\sinh(\pi\omega/2)$. This follows from the beta integral $\int_{\mathbb{R}} e^{-i\omega y}\,\mathrm{sech}^2(y) dy = 2\Gamma(1+\frac{i\omega}{2})\Gamma(1-\frac{i\omega}{2})$ (substitute $u = (1+\tanh y)/2$, so $du = \frac{1}{2}\,\mathrm{sech}^2\, y\, dy$ and $e^{2y} = u/(1-u)$) and $|\Gamma(1+ix)|^2 = \pi x/\sinh(\pi x)$; see [5, Sections 3.98 and 8.33]. The transform is positive, even and exponentially decaying. Fourier inversion and Fubini therefore give, for even $k = m-1$,

$$\mathbb{E}[\tanh^{(m)}(\sigma G)] = \frac{1}{2\pi}\int_{\mathbb{R}} \frac{\pi\omega}{\sinh(\pi\omega/2)} (i\omega)^k e^{-\sigma^2\omega^2/2} d\omega = (-1)^{(m-1)/2} I_m(\sigma),$$

so $c_m = \beta_m^2/v = \sigma^{2m} I_m^2/(m!v)$. For the lower bound use $1/\sinh y \geq 2e^{-y}$. Let $J_m = \int_0^\infty \omega^m e^{-\sigma^2\omega^2/2} d\omega = \sigma^{-m-1} 2^{(m-1)/2}\Gamma(\frac{m+1}{2})$, and let W have density proportional to $\omega^m e^{-\sigma^2\omega^2/2}$

on $(0,\infty)$. By Jensen's inequality, $I_m \ge 2J_m \mathbb{E}e^{-\pi W/2} \ge 2J_m e^{-\pi \mathbb{E}W/2}$. Here $\mathbb{E}W = J_{m+1}/J_m = \sqrt{2}\,\Gamma(\frac{m}{2}+1)/(\sigma\Gamma(\frac{m+1}{2})) \le \sqrt{m+1}/\sigma$, by $\Gamma(x+\frac{1}{2}) \le \sqrt{x}\,\Gamma(x)$. Finally the duplication formula gives $\sigma^{2m}J_m^2/m! = \sigma^{-2}\frac{\sqrt{\pi}}{2}\Gamma(\frac{m+1}{2})/\Gamma(\frac{m}{2}+1) \ge \sigma^{-2}\frac{\sqrt{\pi}}{2}\sqrt{2/(m+1)}$, again by $\Gamma(x+\frac{1}{2}) \le \sqrt{x}\,\Gamma(x)$ with $x = \frac{m+1}{2}$. □

The formula is checked numerically in `numerics/spectrum.py`: the computed coefficients sum to one up to the truncation mass, and they reproduce a direct quadrature of the kernel (Section 7).

Corollary 4.2. *For $\phi = \tanh$ and every depth and gains, the composite coefficients of F satisfy, for every odd m,*

$$c_m \;\ge\; \kappa\, e^{-C_\phi\sqrt{m}}, \qquad \kappa = \frac{2\sqrt{2\pi}}{\sigma_1^2 v_1}\prod_{\ell=2}^{L-1} c_1(q_\ell) > 0, \qquad C_\phi = \sqrt{2}\Big(\frac{\pi}{\sigma_1}+1\Big).$$

Proof. Since every $\widehat{\phi}_{q_\ell}$ has nonnegative coefficients, $\widehat{\phi}_{q_\ell}(u) \ge c_1(q_\ell)u$ coefficientwise, so $[t^m]F \ge \prod_{\ell\ge 2} c_1(q_\ell)\cdot c_m(q_1)$. Here $c_1(q) = q(\mathbb{E}\,\mathrm{sech}^2(\sqrt{q}G))^2/v_\phi(q) > 0$. Apply (5) with $\sigma = \sigma_1$, and use $(m+1)^{-1/2} \ge e^{-\sqrt{m+1}}$ and $\sqrt{m+1} \le \sqrt{2m}$. □

Lemma 4.3 (Eigenvalue lower bounds)**.** *Let F be odd with $c_m \ge \kappa e^{-\psi(m)}$ for every odd m, where $\kappa > 0$ and ψ is nondecreasing. For $d \ge 101$:*

(a) $\binom{d}{q}\lambda_q \;\ge\; \dfrac{A\kappa\, e^{-\psi(40D)}}{2D(D+1)}$ *for every $0 \le q \le 0.49d$;*

(b) the smallest eigenvalue of the matrix K satisfies $v_{\min}(K) \ge \frac{A\kappa}{8}e^{-\psi(\lceil 10D\log(2D)\rceil)}$.

Proof. Poissonize the walk. If the number of steps is $\Pi_t \sim \mathrm{Poisson}(t)$, written Π_t to avoid confusion with the width, the coordinates flip independently, each an odd number of times with probability $p(t) = (1-e^{-2t/D})/2$. Hence $\sum_m \pi_t(m)p_m(j) = p(t)^j(1-p(t))^{D-j}$, where $\pi_t(m) = e^{-t}t^m/m!$. For odd j, only odd m contribute, and $p_m(j) \le 1$ and $\pi_t(m) \le 1$. Hence, for every integer m_1,

$$a_j \;\ge\; \Big(\min_{m\le m_1,\ m\ \text{odd}} c_m\Big)\Big[p(t)^j(1-p(t))^{D-j} - \mathbb{P}\{\Pi_t > m_1\}\Big], \tag{6}$$

and $\mathbb{P}\{\Pi_t > m_1\} \le (et/m_1)^{m_1}$ whenever $et < m_1$, by the Chernoff bound $\mathbb{P}\{\Pi_t \ge m\} \le e^{-t}(et/m)^m$ at $m = m_1+1$.

(a) Let $0 \le q \le 0.49d$ and let $j \in \{q, q+1\}$ be odd. Then $j \le 0.49d+1 \le 0.495D$ because $D \ge 102$. Choose t with $p(t) = j/D$; then $t = \frac{D}{2}\log\frac{D}{D-2j} \le \frac{D}{2}\log 100 < 2.31D$. Take $m_1 = 40D$. Then $\mathbb{P}\{\Pi_t > m_1\} \le 0.158^{40D} < e^{-73D}$. The $\mathrm{Bin}(D, j/D)$ probability of its mode j is at least $1/(D+1)$, so (6) gives $\binom{D}{j}a_j \ge \kappa e^{-\psi(40D)}[\frac{1}{D+1} - 2^D e^{-73D}] \ge \kappa e^{-\psi(40D)}/(2(D+1))$. Since $\lambda_q = Aa_j$ by Lemma 2.5(e), and $\binom{d}{q}/\binom{D}{j} \ge \min\{\frac{D-q}{D}, \frac{q+1}{D}\} \ge \frac{1}{D}$, part (a) follows.

(b) Let $j \le D$ be odd, $t = \frac{D}{2}\log(2D)$ and $m_1 = \lceil 10D\log(2D)\rceil \ge 20t$. Then $p(t)^j(1-p(t))^{D-j} \ge 2^{-D}(1-\frac{1}{2D})^D \ge 2^{-D-1}$, and $\mathbb{P}\{\Pi_t > m_1\} \le (e/20)^{m_1} < 2^{-D}/4$. Thus $a_j \ge \kappa e^{-\psi(m_1)}2^{-D}/4$. Every λ_q equals Aa_j for an odd $j \le D$, and the matrix eigenvalues are $M\lambda_q = 2^{D-1}\lambda_q$. □

5 The margin learner

Let $G \sim N(0,K)$ with K as in (3), F odd, and

$$A_S = \{|G(x)| > a\ \forall x;\ y_iG(X_i) > 0\ \forall i\}, \qquad \mathcal{D}(S,y) = -\log\mathbb{P}(A_S)$$

for labels $y \in \{-1,1\}^n$. The posterior is the law of G given A_S; its relative entropy from $N(0,K)$ is exactly $\mathcal{D}(S,y)$. This is the $\gamma = 0$ structure of [1, Section 7], with the arcsine kernel replaced by K.

5.1 Conditioning cost

The bump functions below are those of [1, Lemma 7.2]; we repeat the parts of that proof that involve the kernel.

Lemma 5.1 (Globally margined interpolation)**.** *Suppose* $\varepsilon_d = \min_{q\le 0.49d}\binom{d}{q}\lambda_q > 0$, *and let* $v_{\min} = v_{\min}(K)$ *(the smallest eigenvalue of the matrix* K*). Put*

$$E_b = \frac{400}{(d+1)\varepsilon_d} + \frac{(d+1)e^{-0.0062d}}{v}.$$

Let $X_1,\dots,X_n$ *be independent and uniform, with* $n(0.2)^{0.1d} \le 0.04$. *With probability at least* $1 - n^2e^{-0.19d} - n\sqrt{A\,0.72^d/v_{\min}}$, *the following holds simultaneously for every labeling* y *and every* $A' > 0$. *There is* $h : \mathcal{X} \to \mathbb{R}$ *with* $|h(x)| > A'$ *for all* x, $y_i h(X_i) > A'$ *for all* i, *and*

$$\|h\|_K^2 \le 8A'^2\Big(\frac{1}{\lambda_0} + 8nE_b\Big).$$

Proof. For $z \in \mathcal{X}$ let $b_z(x) = 20\int_{2/5}^{9/20}(1-2p)^{\mathrm{dist}(x,z)}dp$. Then $b_z(z) = 1$, $0 \le b_z(x) \le (1/5)^{\mathrm{dist}(x,z)}$, and $\widehat{b_z}(U) = \chi_U(z)\widehat{b}_{|U|}$ with $\widehat{b}_j = 20\int_{2/5}^{9/20}(1-p)^{d-j}p^j dp$. Hence $\binom{d}{j}\widehat{b}_j \le 20/(d+1)$ and $\widehat{b}_j \le (3/5)^d$, and for $j > 0.49d$ the integrand increases in p, so $\widehat{b}_j \le 0.55^{d-j}0.45^j$ [1, proof of Lemma 7.2]. Therefore

$$\|b_z\|_K^2 = \sum_j \binom{d}{j}\frac{\widehat{b}_j^2}{\lambda_j} \le \sum_{j\le 0.49d}\frac{(20/(d+1))^2}{\varepsilon_d} + \frac{2^d}{v}\sum_{j>0.49d}\binom{d}{j}0.55^{2(d-j)}0.45^{2j} \le E_b,$$

using $\lambda_j \ge v_{\min}/M$. For the last sum, the exponent $\log 2 + \mathsf{h}(t) + 2(1-t)\log 0.55 + 2t\log 0.45$, with h the binary entropy in nats, decreases for $t \ge 0.49$ and is below -0.0062 at $t = 0.49$, as computed in [1, proof of Lemma 7.2].

Let $J(z,w) = \langle b_z, b_w\rangle_K = \sum_U \nu_{|U|}\chi_U(z)\chi_U(w)$, with $\nu_j = \widehat{b}_j^2/\lambda_j \le 0.36^d M/v_{\min} = 0.72^d/v_{\min}$. For independent uniform X, Y, Parseval gives $\mathbb{E}J(X,Y)^2 = \sum_j \binom{d}{j}\nu_j^2 \le \nu_{\max}\|b_z\|_K^2$. Since $E_b \ge \|b_z\|_K^2 \ge b_z(z)^2/K(z,z) = 1/A$, Markov's inequality gives $\mathbb{P}\{\sum_{i\ne j}|J(X_i,X_j)| > nE_b\} \le n\sqrt{\nu_{\max}/E_b} \le n\sqrt{A\,0.72^d/v_{\min}}$. On the complement, every subset T of the indices satisfies $\|\sum_{i\in T} b_{X_i}\|_K^2 \le |T|E_b + \sum_{i\ne j}|J| \le 2nE_b$.

With probability at least $1-n^2e^{-0.19d}$ all pairwise distances exceed $0.2d$, since $\log 2 - \mathsf{h}(0.2) > 0.19$. Then each x has at most one sample point within distance $0.1d$, so $\sum_{i\in T} b_{X_i}(x) \le 1 + n(0.2)^{0.1d}$, and the sum is at most $0.2 + 0.04$ unless $x = X_i$ with $i \in T$, where it is at least 1. Take $T = \{i : y_i = -1\}$ and $h = 2A'(1 - 2\sum_{i\in T} b_{X_i})$. Then $h \le -2A'$ on T and $h \ge 1.04A'$ elsewhere. Finally $\|h\|_K^2 \le 8A'^2(\|1\|_K^2 + 4\|\sum_T b\|_K^2)$, and $\|1\|_K^2 = 1/\lambda_0$. □

Lemma 5.2 (Conditioning cost)**.** *On the event of Lemma 5.1, for every labeling,*

$$\mathcal{D}(S,y) \le \log 2 + 4A'^2\Big(\frac{1}{\lambda_0} + 8nE_b\Big), \qquad A' = a + \sqrt{2A\log(4M)}.$$

Proof. Let $t_0 = \sqrt{2A\log(4M)}$ and $B = \{\max_x |G(x)| \le t_0\}$. A Gaussian tail and union bound give $\mathbb{P}(B) \ge 1/2$. With h from Lemma 5.1 at level $A' = a + t_0$, $h + B \subseteq A_S$. For a symmetric set B, the Cameron–Martin formula and $\cosh \ge 1$ give $\mathbb{P}(G \in h + B) \ge \mathbb{P}(B)e^{-\|h\|_K^2/2}$, as in [1, proof of Lemma 7.1]. □

5.2 Gibbs error

Lemma 5.3 (Gaussian quadratic entropy [1, Lemma 9.1]). *If Z is a linear image of a centered Gaussian vector, with covariance of trace t and largest eigenvalue ℓ, then $\mathbb{E}_Q\|Z\|^2 \le 2t + 4\ell D_0$ for every law Q of the underlying vector whose relative entropy from the Gaussian law is at most D_0.*

Proposition 5.4 (Gibbs error for every target). *Let K be any positive-definite covariance on $\mathcal{X}$ that depends only on Hamming distance and has nonincreasing degree eigenvalues. For every target f, sample S and $0 \le s < d$, the posterior given A_S (labels $y_i = f(X_i)$) satisfies*

$$R_S := \mathbb{E}\big[\mathbb{P}_x\{f(x)G(x) < 0\} \mid A_S\big] \;\ge\; \mathrm{err}_s(f) - \frac{2\tau_s + 4\lambda_{s+1}\mathcal{D}(S,f)}{a^2}.$$

Proof. Apply Lemma 5.3 to $Z = M^{-1/2}G_{>s}$, whose covariance has trace τ_s and largest eigenvalue λ_{s+1}. This gives $\mathbb{E}[\|G_{>s}\|_2^2 \mid A_S] \le 2\tau_s + 4\lambda_{s+1}\mathcal{D}$. Fix a posterior draw and let $p = G_{\le s}$. Where $f(x)p(x) \le 0 < f(x)G(x)$ we have $|G - p| \ge fG = |G| > a$. Hence $\mathbb{P}_x\{fp \le 0\} \le \mathbb{P}_x\{fG < 0\} + \|G_{>s}\|_2^2/a^2$, and the left side is at least $\mathrm{err}_s(f)$. Take the posterior expectation. □

Theorem 5.5 (Margin learner, tanh). *Let $\phi = \tanh$, and fix $L \ge 2$, gains $\sigma_1, \dots, \sigma_L > 0$ and a margin $a > 0$. There is d_0 such that for every $d \ge d_0$ and every integer $0 \le n \le \exp(\frac{1}{20}\sqrt{d}\log d)$ the following holds. With probability at least $1 - e^{-d/10}$ over independent uniform training inputs, the margin learner for full parity $f = \chi_d$ satisfies*

$$\lim_{N\to\infty} R_N(S) \ge \frac{3}{8}, \qquad \liminf_{N\to\infty} F_N(S) \ge \frac{1}{16}.$$

More generally, there is C, depending on L, the gains and a, such that on the same sample event, for every target f and every odd $s < d$,

$$\lim_{N\to\infty} R_N(S) \;\ge\; \mathrm{err}_s(f) - \frac{2A\,T(s)}{a^2}\Big(1 + \frac{2C(d^2 + n\,e^{C\sqrt{d}})}{\binom{d+1}{s+2}}\Big). \tag{7}$$

Proof. Spectral inputs. By Corollary 4.2, Lemma 4.3 applies with $\psi(m) = C_\phi\sqrt{m}$. Thus $\varepsilon_d \ge e^{-C\sqrt{d}}$ and $v_{\min} \ge e^{-C\sqrt{d\log d}}$ for large d, where here and below C denotes constants depending only on L, the gains and a. In particular $E_b \le e^{C\sqrt{d}}$. Also $\lambda_0 = Aa_1 \ge Ac_1/D$, since $p_1(1) = 1/D$.

Sample event. Let $n \le \exp(\frac{1}{20}\sqrt{d}\log d) = e^{o(d)}$. Then $n(0.2)^{0.1d} \le 0.04$, and the failure probability in Lemma 5.1 is at most $e^{-0.18d} + e^{-0.16d + C\sqrt{d\log d}} \le e^{-d/10}$ for large d. On this event Lemma 5.2 and $A'^2 \le Cd$ give, for every labeling, $\mathcal{D}(S,y) \le C(d^2 + ne^{C\sqrt{d}})$.

Every target. Combine Proposition 5.4 with Lemma 2.6(b),(c), which give $\lambda_{s+1} \le \tau_s/\binom{D}{s+2}$ and $\tau_s \le AT(s)$. Then Proposition 2.4(iii) gives (7).

Parity. Let s be the largest odd integer with $s+1 \le \sqrt{d}/5$. Lemma 3.3 gives $\mathrm{err}_s(\chi_d) > 0.42$. Since $s+2 \ge \sqrt{d}/5 - 1$, $\log\binom{D}{s+2} \ge (s+2)\log\frac{D}{s+2} \ge \frac{1}{10}\sqrt{d}\log d - \log d$. For $n \le \exp(\frac{1}{20}\sqrt{d}\log d)$ the fraction in (7) is therefore at most $\exp(-\frac{1}{20}\sqrt{d}\log d + C\sqrt{d} + C\log d) \to 0$. Also $T(s) \to 0$ as $d \to \infty$, because $s \to \infty$ and T does not depend on d (Lemma 2.3). Hence $\lim_N R_N(S) \ge 0.42 - o(1) \ge 3/8$ for large d, and (2) gives the bound on F_N. □

Remark 5.6 (Thresholds, constants, activations). (1) Taking $s+1 \le \delta\sqrt{d}$ in the proof gives $\lim_N R_N \ge \frac{1}{2} - \frac{\delta}{\sqrt{2\pi}} - o(1)$ for $n \le \exp(\frac{\delta}{4}\sqrt{d}\log d)$. For $q > 1/2$ take any $\delta < \sqrt{2\pi}(q - \frac{1}{2})$ and $c_q = \delta/4$; the inequality below holds at every width and passes to the limit since $R_N \to R_S$. By the second inequality in FAQ Question 5, $F_q \ge (R - (1-q))/q$, so for each fixed $q > 1/2$ the fraction of inputs

with correct-label probability at most q stays bounded below in a range $n \le \exp(c_q\sqrt{d}\log d)$. (2) The constant $\frac{1}{20}$ is not optimized. (3) The proof uses tanh only through oddness, Assumption 2.1, and the lower bound of Corollary 4.2. It works verbatim for any odd activation satisfying Assumption 2.1 whose composite coefficients satisfy $c_m \ge \kappa e^{-\psi(m)}$ for odd m (by the argument of Corollary 4.2 it suffices that the first-layer coefficients satisfy this and that $c_1(q_\ell) > 0$ for $\ell \ge 2$), with ψ nondecreasing and $\psi(m) = o(\sqrt{m}\log m)$. Then $E_b = e^{o(\sqrt{d}\log d)}$, $v_{\min} \ge e^{-o(d)}$, and the same threshold results. The error function is not covered: since $\mathbb{E}\,\mathrm{erf}(\sigma U)\,\mathrm{erf}(\sigma V) = \frac{2}{\pi}\arcsin(\frac{2\sigma^2\rho}{1+2\sigma^2})$, its dual coefficients decay geometrically in m. (4) For k-sparse parity the same proof needs $e^{C\sqrt{d}} \ll \binom{D}{\delta\sqrt{k}}$, roughly $k \ge Cd/\log^2 d$; we have not pursued smaller k.

6 Where the exponential proof breaks

The $\gamma = 0$ part of the subcritical theorem [1, Theorem 9.2] covers n up to e^{cd}. It chooses an adaptive odd cutoff r with $n \approx A_\delta D_r \log D_r$, where $D_r = \sum_{q\le r}\binom{d}{q}$. Its key normalized estimate is

$$\frac{\mathcal{D}(S)}{n} \le Cd\rho^{-\beta}, \qquad \rho = r/d, \qquad \text{[1, eq. (42)]},$$

a bound polynomial in d (at most $Cd^{1+\beta}$). It combines this with the head bound $\mathbb{E}^*\|G_{\le r}\|_2^2 \le 8\tau_r + 16 b_r\tau_r\mathcal{D}/n$ [1, eq. (45)], where $\mathbb{E}^*$ is the posterior expectation and b_r bounds the sampled Gram matrix of the tail kernel. The clipped parity lemma is then applied at the scale of the posterior output amplitude, which is of order $d^{(1-\beta)/2}$ there [1, Lemma 7.5]. That amplitude step uses the variance $\tau_u \asymp d^{-\beta}$ of the Fourier levels above $u \approx d/50$; at the standard scaling this variance is tiny, and Lemma 7.5 gives no useful amplitude. We show below that the normalization (42) also fails: the conditioning cost per sample is superpolynomial in d. We then explain what estimate would let the rest of the argument go through.

Fix $\delta \in (0, 1/8)$, let A_δ be the sampling constant of [1, Lemma 4.4] (squared coercivity: with probability at least $1-\delta$, $\frac{1}{n}\sum_i (f(X_i)P(X_i))_-^2 \ge \frac{1}{4}\|P\|_2^2$ for all P of degree at most r, if $r \le d/64$ and $n \ge A_\delta D_r \log(2D_r)$), and let $r = r(n)$ be the largest odd integer with $A_\delta D_r \log(2D_r) \le n$.

Lemma 6.1 (Universal tail-Gram bound)**.** *Let F be odd. There are $c_1 > 0$ and C_δ, depending only on δ and A_δ, such that for large d, every n with $\exp((\log(d+1))^3) \le n \le e^{c_1 d}$, and $r = r(n)$, the following holds. With probability at least $1-\delta$ over iid uniform inputs, the matrix $Q_{SS} = ((K_{>r})(X_i, X_j))_{i,j\le n}$ satisfies*

$$\|Q_{SS}\|_{\mathrm{op}} \le b_r\tau_r, \qquad b_r = C_\delta(1 + \sqrt{d\,rH_r} + rH_r), \qquad H_r = \log(eD/r),$$

and $r \le d/64$, $r \ge (\log d)^2/4$.

Proof. The range of r follows exactly as in [1, Lemma 6.4], which uses only binomial estimates. That lemma's argument also gives $n < A_\delta D_{r+2}\log(2D_{r+2}) \le CA_\delta\binom{D}{r+2} rH_r$. [1, Lemma 5.2] bounds $\mathbb{P}\{\|Q_{SS} - \tau I\| > z\tau\}$ in terms of four quantities. Its proof uses only these properties of Q: positive semidefinite, constant diagonal $\tau = \tau_r$, $\max_x \sum_y Q(x,y)^2 = Ms_{2,r}$, $\|Q\|_{\mathrm{op}} = M\lambda_{r+1}$, and the far-pair bound $\eta_r = \max_{d/4\le \mathrm{dist}\le 3d/4}|Q(x,y)|$. Here

$$\frac{n\lambda_{r+1}}{\tau_r} \le \frac{n}{\binom{D}{r+2}} \le CA_\delta rH_r, \qquad \frac{\sqrt{n\cdot 2\log M\cdot s_{2,r}}}{\tau_r} \le \sqrt{2d\log 2\,\frac{n}{\binom{D}{r+2}}} \le C\sqrt{A_\delta d\,rH_r},$$

by Lemma 2.6(b),(d). For the far pairs, the proof of [1, Lemma 5.1] gives $\eta_r \le \tau_r(C\sqrt{D}e^{-r/8} + e^{-c_0 D})$, once two of its steps are reread. First, its bound for $r < j \le D - r$ uses only the Krawtchouk estimate

(20) of that paper together with $\sum_{j>r}\binom{D}{j}a_j = \tau_r/A$ (here $Q = \sum_{j\geq r+2,\ j \text{ odd}} Aa_j\mathsf{K}_j$, because F is odd and r is odd). Second, for $j > D - r$ it uses monotonicity: $a_j \leq a_b \leq \tau_r/(A\binom{D}{b})$ for an odd b near $D/4$. Thus $2\log M \cdot \eta_r/\tau_r \to 0$ for $r \geq (\log d)^2/4$. The near-pair and collision terms are at most $3ne^{d\mathsf{h}(1/4)}/2^d$ and $n^2 \cdot 3e^{d\mathsf{h}(1/4)}/2^d$. They vanish when $2c_1 < \log 2 - \mathsf{h}(1/4)$. Choosing z as a δ-dependent multiple of the sum of the four terms gives the claim. □

Proposition 6.2 (The conditioning cost per sample is superpolynomial)**.** *Let $\phi = \tanh$; fix L, gains, a and $\delta \in (0, 1/8)$. For large d and every n with $\exp((\log(d+1))^3) \leq n \leq e^{c_1 d}$, with probability at least $1 - 2\delta$ over iid uniform inputs labelled by full parity,*

$$\frac{\mathcal{D}(S)}{n} \;\geq\; \frac{a^2 - 10\tau_r}{\tau_r(16b_r + 4CA_\delta r H_r)}, \qquad r = r(n).$$

Consequently, for every $\epsilon > 0$ and k, for all large d and every n with $\exp(d^\epsilon) \leq n \leq e^{c_1 d}$, on the same event, $\mathcal{D}(S)/n \geq d^k$. In the other direction, on the event of Lemma 5.1, $\mathcal{D}(S)/n \leq e^{C\sqrt{d}}$ for every $n \geq 1$. (Some upper limit on n is necessary in the first statement: for fixed d, $\mathcal{D}(S)$ is bounded by the cost of the full sign pattern of f, so $\mathcal{D}(S)/n \to 0$ as $n \to \infty$.)

Proof. Work on the intersection of the squared-coercivity event and the event of Lemma 6.1. Write $p = G_{\leq r}$ and $\mathbb{E}^*$ for the posterior expectation. Every posterior draw has $|G| > a$ everywhere, so $\mathbb{E}^*\|G\|_2^2 \geq a^2$. At the training inputs, $f(X_i)G(X_i) > 0$ implies $(f(X_i)p(X_i))_- \leq |G_{>r}(X_i)|$. Coercivity gives $\|p\|_2^2 \leq \frac{4}{n}\sum_i G_{>r}(X_i)^2$. Lemma 5.3, applied to $n^{-1/2}(G_{>r}(X_i))_i$, whose covariance Q_{SS}/n has trace τ_r and largest eigenvalue at most $b_r\tau_r/n$, gives $\mathbb{E}^*\frac{1}{n}\sum_i G_{>r}(X_i)^2 \leq 2\tau_r + 4b_r\tau_r\mathcal{D}/n$. Applied to $M^{-1/2}G_{>r}$ it gives $\mathbb{E}^*\|G_{>r}\|_2^2 \leq 2\tau_r + 4\lambda_{r+1}\mathcal{D}$, with $\lambda_{r+1} \leq \tau_r/\binom{D}{r+2} \leq \tau_r CA_\delta r H_r/n$. Hence

$$a^2 \leq \mathbb{E}^*\|p\|_2^2 + \mathbb{E}^*\|G_{>r}\|_2^2 \leq 10\tau_r + \frac{\tau_r\mathcal{D}}{n}(16b_r + 4CA_\delta r H_r),$$

which is the first claim. For the second, the tanh dual activations have all moments finite: if $h(g) = \tanh(\sqrt{q}g)$, the identity $\mathbb{E}[h^{(k)}(G)He_j(G)] = \mathbb{E}[h(G)He_{j+k}(G)]$ gives $\sum_m \frac{m!}{(m-k)!}\beta_m^2 = \mathbb{E}[h^{(k)}(G)^2] < \infty$. Regard (c_m) as a probability distribution on degrees. For a composition $\widehat{\phi} \circ H$, the degree distribution is the law of a random sum $\sum_{i\leq Y} J_i$, where Y has the distribution of the coefficients of $\widehat{\phi}$ and the J_i are independent copies of the degree under H, independent of Y. Since $(\sum_{i\leq Y} J_i)^k \leq Y^{k-1}\sum_{i\leq Y} J_i^k$, we get $\mathbb{E}(\sum_{i\leq Y} J_i)^k \leq \mathbb{E}Y^k\,\mathbb{E}J^k$. So $T(r) \leq C_k r^{-k}$ for every k, and $\tau_r \leq AT(r)$. If $n \geq \exp(d^\epsilon)$, then $D_{r+2}\log(2D_{r+2}) > n/A_\delta$ forces $r \geq d^{\epsilon/2}$ for large d, so $\tau_r \leq d^{-k'}$ for every k', while $b_r + rH_r \leq Cd\log d$. The upper bound follows from Lemma 5.2 and $E_b \leq e^{C\sqrt{d}}$. □

What this shows and does not show. The proposition is about a proof technique, not about sample complexity, and it refutes only the normalization (42). It does not show that the subcritical strategy fails. The clipped-parity lemma at scale a only needs the posterior's low-degree part to be $O(a)$ in L^2, not small, because it is multiplied by $u/d \approx \rho$, and ρ can be chosen small. The following is an *extension suggested by the proof*, not proved here. Suppose that on a sample event of probability at least $3/4$ there were a matching upper bound

$$\frac{\mathcal{D}(S)}{n} \leq \frac{K\,a^2}{\tau_r\,(b_r + rH_r)}, \tag{8}$$

with K independent of d, n and ρ. Work on the intersection of that event with the coercivity event and the event of Lemma 6.1 (probability at least $3/4 - 2\delta$), for $\exp((\log(d+1))^3) \leq n \leq e^{c_1 d}$, and

take $u = r+2\lceil 10\log(d+1)\rceil$. The head bound would then give $\mathbb{E}^*\|G_{\le r}\|_2^2 \le 8\tau_r + 16Ka^2$. Lemma 5.3 with $\lambda_{r+1} \le \tau_r/\binom{D}{r+2}$ and $n/\binom{D}{r+2} \le CA_\delta r H_r$ would bound the band $(r,u]$ by $2\tau_r + CKa^2$. The same lemma with $\lambda_{u+1} \le \tau_r/\binom{D}{u+2}$ and the geometric decay of $\binom{D}{r+2}/\binom{D}{u+2}$ would make the tail above u $o(1)$: it is at most $2\tau_r + 4Ka^2\, n/(\binom{D}{u+2}(b_r + rH_r)) \le 2\tau_r + CKa^2\binom{D}{r+2}/\binom{D}{u+2}$, and the binomial ratio is at most d^{-10} for $r \le \alpha d$ with α small, as in [1, Lemma 6.4]. The clipped-parity lemma would then give $|\mathbb{E}^*\mathbb{E}_x f \operatorname{sign} G| \le C\rho\sqrt{K} + o(1)$, which is an $\exp(\Omega(d))$ bound once $\rho \le \alpha$ is small enough. Proposition 6.2 says that (8) cannot hold with a smaller right-hand side by more than a constant factor. So the missing ingredient is an upper bound on the conditioning cost that matches the proposition up to constants. The only upper bound we have is the bump bound $e^{C\sqrt{d}}$.

A second sufficient route avoids the adaptive cutoff. It would suffice to show that, for a small fixed α,

$$\mathbb{E}^*\|G_{\le \alpha d}\|_2^2 \le C \quad \text{and} \quad \mathbb{E}^*\|G_{>\alpha d}\|_2^2 = o(1),$$

with C independent of α. By the clipped-parity lemma at scale a, $|\mathbb{E}_x f \operatorname{sign} G| \le \frac{\alpha}{a}\|G_{\le \alpha d}\|_2 + \|G_{>\alpha d}\|_2/a$, so it is enough that $\alpha\sqrt{C}/a \le 1/4 - o(1)$. The second condition holds for $n \le e^{c_\alpha d}$ with $c_\alpha < \min\{\mathsf{h}(\alpha)/2, 0.09\}$, on the event of Lemma 5.1, which then has probability at least $1 - e^{-c'd}$. Indeed Lemma 5.3 gives $\mathbb{E}^*\|G_{>s}\|_2^2 \le 2\tau_s + 4\lambda_{s+1}\mathcal{D}$, and Lemmas 2.6 and 5.2 make the right side $O(T(s)) + e^{C\sqrt{d}} n/\binom{D}{s+2}$ at odd $s \approx \alpha d$. The first condition is an upper bound on the posterior's low-degree amplitude, and we do not know how to prove it. The entropy bound gives only $\mathbb{E}^*\|G\|_2^2 \le 2A + 4\lambda_0\mathcal{D} = O(1 + \mathcal{D}/d)$, which is $O(1)$ only if $\mathcal{D} = O(d)$; no bound here provides that.

Heuristic picture (heuristic; not used above). For $\exp(d^\epsilon) \le n \le e^{c_1 d}$, the proposition and the bump construction place the conditioning cost per sample between $d^{\omega(1)}$ and $e^{C\sqrt{d}}$. The mechanism behind the lower bound is coercivity. At the training inputs parity cannot be followed by a low-degree polynomial, so the low-degree part is dominated by the high-degree part there. Yet the global margin forces $\|G\|_2 \ge a$. So the posterior must place L^2 mass of order a^2 above degree r, where the prior variance τ_r is tiny. If that mass sits mainly in the band just above r, the cost should match the proposition, and the argument above would give $\exp(\Omega(d))$. If instead the posterior used large-amplitude low-degree thresholds that approximate parity, the bound could fail. We have not ruled this out, and the numerics below are too small to test it.

7 Numerical illustrations

All scripts and outputs are in `numerics/` next to this paper. They are illustrations, not evidence about asymptotic exponents.

Spectra (`spectrum.py`, `spectrum.txt`). We compute the composite coefficients c_m from Lemma 4.1 for the unit-gain network at $L = 2, 3, 4$ and for the specified-gain network at $L = 3$, for $m \le 600$. We then compute the cube eigenvalues through the walk representation $a_j = \sum_m c_m p_m(j)$, which involves no cancellation, at $d = 20, 40, 80, 160$. At $d = 10$ the eigenvalues agree with a direct Gauss–Hermite evaluation of the layer recursion followed by a Krawtchouk transform. The checks confirm positivity, monotonicity and pairing (Lemma 2.5), and the inequalities of Lemma 2.6(b),(c) and (5). Over the computed range, the level masses $\binom{d}{q}\lambda_q$ of the tanh kernels decay much faster than those of the arcsine kernel, whose level masses decay like $q^{-3/2}$ at $L = 2$ [1, Lemma 3.2]. Table 1 records representative values.

kernel	d	$q=1$	$q=3$	$q=9$	$q=17$	$q=33$	$q=65$	$v_{\min}$
tanh unit gain $L=2$	80	-0.04	-1.28	-3.26	-5.20	-8.86	-22.30	-15.3
tanh unit gain $L=2$	160	-0.03	-1.26	-3.19	-4.95	-7.77	-13.18	-23.3
tanh unit gain $L=3$	80	-0.05	-1.15	-3.06	-5.04	-8.78	-22.25	-15.3
tanh unit gain $L=3$	160	-0.05	-1.14	-2.98	-4.78	-7.67	-13.13	-23.3
tanh specified gain $L=3$	80	-0.07	-1.03	-2.67	-4.38	-7.82	-20.95	-13.9
tanh specified gain $L=3$	160	-0.06	-1.02	-2.60	-4.15	-6.76	-11.89	-21.3
arcsine $L=2$	80	-0.20	-0.98	-1.74	-2.17	–	–	-1.8
arcsine $L=2$	160	-0.20	-0.98	-1.73	-2.15	-2.60	–	-2.4

Table 1: $\log_{10}$ of the level mass $\binom{d}{q}\lambda_q/A$ of the normalized kernel $F((1+x\cdot y)/(d+1))$, and $\log_{10}$ of the smallest eigenvalue of the matrix $F((1+x\cdot y)/(d+1))_{x,y}$. Unit gain means $\sigma_\ell=1$ for all ℓ; specified gain is the family of [2, Section 7] at $L=3$. The computation truncates the slowly decaying arcsine coefficients at $m=600$ (omitted mass 0.021), so arcsine entries are shown only where the omitted terms are negligible ($q\le 17$ at $d=80$, $q\le 33$ at $d=160$), and the arcsine $v_{\min}$ is only a lower bound. All checks listed in Section 7 passed for every row.

Sign-only posteriors (`signgp_parity.py`, `signgp_parity.txt`). We run exact Hamiltonian Monte Carlo for the orthant-truncated Gaussian, using the validated sampler of [3], for full parity at $d=10,12$. The kernels are tanh unit gain at $L=2,3$, tanh specified gain at $L=3$, and the arcsine kernel at $L=2$. Table 2 gives the confidence-failure fraction. In this range all four kernels fail on most unseen inputs (at least 71% of them in every cell). The whole-cube fraction falls mainly because more inputs are observed. The tanh kernels fail somewhat less often than the arcsine kernel, and the three tanh kernels are nearly indistinguishable. At these dimensions the numbers say nothing about the exponent.

$d=10$, $n=100,200,400,800$				
tanh unit gain, $L=2$	0.71 (0.78)	0.62 (0.75)	0.53 (0.79)	0.41 (0.90)
tanh unit gain, $L=3$	0.71 (0.78)	0.62 (0.76)	0.54 (0.79)	0.41 (0.90)
tanh specified gain, $L=3$	0.71 (0.79)	0.64 (0.78)	0.55 (0.81)	0.42 (0.92)
arcsine, $L=2$	0.77 (0.85)	0.75 (0.91)	0.64 (0.94)	0.45 (0.98)
$d=12$, $n=200,400,800,1600$				
tanh unit gain, $L=2$	0.71 (0.74)	0.65 (0.71)	0.61 (0.74)	0.55 (0.82)
tanh unit gain, $L=3$	0.71 (0.75)	0.65 (0.72)	0.62 (0.75)	0.55 (0.82)
tanh specified gain, $L=3$	0.72 (0.76)	0.67 (0.74)	0.63 (0.77)	0.56 (0.83)
arcsine, $L=2$	0.82 (0.87)	0.80 (0.88)	0.76 (0.92)	0.64 (0.95)

Table 2: Sign-only learner for full parity: confidence-failure fraction $F=\mathbb{P}_x\{q_S(x)\le 2/3\}$ over the whole cube (observed inputs count as successes), and in parentheses over unseen inputs only; mean over 3 seeds of iid uniform training inputs, columns in the order of the listed n. Two exact-HMC chains of 1500 draws after 300 burn-in; the largest difference in F between chains over all runs is 0.006.

8 Status summary

Statement	Status
Width limit to a nondegenerate GP, both learners	Proved (Prop. 2.4)
Sign-only learner, every training set, every activation/depth/gain: $\exp(\Omega(\sqrt{d}\log d))$ for full parity, d^k for fixed k-sparse parity	Proved (Thm. 3.2, Cor. 3.4)
Margin learner with readout exponent $b < 1$, tuple $(0, 1, \ldots, 1, b)$: same bounds	Proved (Remark 3.5)
Margin learner, tanh, typical samples: $\exp(\frac{1}{20}\sqrt{d}\log d)$	Proved (Thm. 5.5)
Same for odd activations with $c_m \geq e^{-o(\sqrt{m}\log m)}$	Proved (Remark 5.6)
Per-sample conditioning cost superpolynomial for $e^{d^\epsilon} \leq n \leq e^{c_1 d}$ (w.p. $\geq 1 - 2\delta$); the subcritical normalization (42) fails	Proved (Prop. 6.2)
A matching upper bound on that cost would give $\exp(\Omega(d))$	Extension suggested by the proof (Section 6)
$\exp(\Omega(d))$ for either learner at fixed depth	Open
Which of the two posterior behaviours in Section 6 occurs	Open (heuristic discussion only)
Margin learner for erf, or for sparse parity with $k \ll d/\log^2 d$	Open

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